



Assessment Task No. 1			
Topic:	Propositional Logic – Syntax, Semantics, and Truth Tables	Week No.	2
Course Code:	CSST101	Term:	1 st Semester
Course Title:	Advance Knowledge Representation and Reasoning	Academic Year:	2025-2026
Student Name	Sarmiento, Janvher Lucas J.	Section	BSCS-3A
Due date		Points	

Propositional Logic Activity Puzzles

Puzzle 1: The Broken Lamp

Propositions:

- P = The lamp is plugged in.
- Q = The bulb is working.
- R = The lamp gives light.

Rules:

1. $(P \wedge Q) \rightarrow R$
2. $\neg R$

Question: What can we conclude about P and Q

Either the lamp is not plugged in ($\neg P$) or the bulb is not working ($\neg Q$), or possibly both

Puzzle 2: The Late Bus

Propositions:

- P = The bus arrived on time.
- Q = The students arrived on time.
- R = The exam started on time.

Rules:

1. $\neg P \rightarrow \neg Q$
2. $\neg Q \rightarrow \neg R$
3. $\neg R$

Question: Did the bus arrive on time?

This doesn't have a definite answer since from $\neg Q \rightarrow \neg R$, knowing $\neg R$ does not allow us to infer $\neg Q$ (that would be affirming the consequent). The chain only gives $R \rightarrow Q \rightarrow P$ (contrapositives), but $\neg R$ gives no information about P . So P is unknown, therefore my answer to this would be undefined.



Puzzle 3: The Locked Door

Propositions:

- P = The key is available.
- Q = The door is unlocked.
- R = The students enter the classroom.

Rules:

1. $P \rightarrow Q$
2. $Q \rightarrow R$
3. $\neg R$

Question: Was the key available?

The key was not available

Puzzle 4: The Basketball Game

Propositions:

- P = The gym lights are working.
- Q = The referee is present.
- R = The basketball game takes place.

Rules:

1. $(P \wedge Q) \rightarrow R$
2. $\neg R$

Question: What condition(s) failed?

At least one condition failed

Puzzle 5: The Forgotten Password

Propositions:

- P = The user remembers the password.
- Q = The user logs into the system.
- R = The user submits the assignment online.

Rules:

1. $P \rightarrow Q$
2. $Q \rightarrow R$
3. $\neg R$

Question: Did the user remember the password?

No, the user did not remember the password